

COMP3821/9801

EXTENDED ALGORITHM DESIGN AND ANALYSIS

# Project 2026

*Proposal Specification*



**DUE**

Week 3

**PROJECT WEIGHTING**

15%

**MARKS AVAILABLE**

20 Marks

## 01 Purpose

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The proposal establishes the initial direction of your project. Explain what your group intends to investigate, communicate, create, or innovate. Explain why it is worthwhile and how you will develop it within the term. Your proposal should give your mentor enough information to provide useful feedback on the idea, its scope, and your planned approach.

At this stage, we assess your objective, evidence of initial understanding, and a realistic plan. Begin engaging with the relevant background so that you can explain the algorithmic ideas your project will involve. New research results, completed proofs, experimental findings and finished resources are not required for the proposal.

## 02 Submission

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Read this brief alongside the *Project Specification*, which describes the project formats, mentoring arrangements and course policies.

|                       |                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Submit</b>         | One .pdf per group.                                                                                                                     |
| <b>Length</b>         | At most three (3) pages of main content, excluding references and AI assistance statements.                                             |
| <b>Identification</b> | Include the project title, intended format(s), each member's name and zID, and your mentor's name.                                      |
| <b>Presentation</b>   | Use a readable font of at least 11pt and margins of at least 2cm. The proposal should be understandable without opening external links. |
| <b>Deadline</b>       | Week 3. The actual deadline will be posted on Moodle.                                                                                   |

## 03 What your proposal should contain

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Address the five areas below. You may use another structure if the required information is easy to find. Write for a reader familiar with COMP3821/9801 who has not studied your topic.

### 1. Topic and objective.

State the central question or purpose precisely. Introduce the problem, relevant assumptions and intended audience. Explain why the topic interests your group and what makes it a worthwhile project.

Make the focus clear enough to guide the work. For example, a scholarly project on maximum flow could compare the invariants and correctness arguments of augmenting-path and push-relabel methods. The topic *maximum flow algorithms* would need this additional focus.

### 2. Initial background and understanding.

Identify relevant sources, algorithms, or existing work, and explain how they inform your direction.

A short and purposeful selection of sources is sufficient; there is no required number of references. Include a brief, concrete explanation that demonstrates initial engagement: an idea, worked example, comparison between methods, or a mathematical difficulty your resource will explain. Give enough detail to show what *you* understand. State what you will still need to learn and distinguish established results from questions you intend to investigate.

**3. Format, scope, and intended outcomes.**

State the project format(s) and what you intend to produce. Explain the substantial algorithmic work involved and what the completed project should demonstrate. Every format includes the final report and poster; identify any additional resource, implementation, or creative work you expect to develop.

Distinguish between a manageable core set of ideas from possible extensions. The core should provide a worthwhile project even if a more ambitious direction proves unsuccessful. Set clear boundaries on the problems, results, or features you will address.

**4. Proposed approaches and evaluations.**

Explain how you will pursue the objective and why the approach is reasonable. Identify the main methods and arguments, comparisons, experiments, or design activities you expect to use. Describe how you will assess correctness and the quality of the intended outcomes, using the guidance of your format in the next section.

Identify any significant uncertainties or dependencies, such as unfamiliar background, an unproven claim, access to suitable inputs or implementation time. Explain a realistic response: maybe a restricted setting, an alternative method, a smaller resource or a revised question. You do not need to resolve these difficulties before submitting.

**5. Plan and collaboration.**

Give milestones leading to the Week 7 progression review and Week 10 poster and final report. Each should identify a concrete outcome: a more developed argument, comparisons between methods studied, implementations, or a resource ready for a walkthrough.

Explain how your group will allocate work, review contributions and develop a shared understanding. Identify immediate next steps and questions for your mentor. A compact table is sufficient.

## 04 Adapting the proposal to your format

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All formats should demonstrate substantial algorithmic understanding and careful reasoning. Use the following table below to guide your proposal's work and how you evaluate your work. These are planning expectations for the proposal.

| Format       | What to make clear in the proposal?                                                                                                                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theoretical  | Define the mathematical question and any restricted setting you plan to study. Identify the techniques or intermediate claims you will explore, as well as the correctness, complexity or other arguments the work will require. If the main question is ambitious, describe a useful set of core problems you plan to explore.                      |
| Scholarly    | Identify a focused set of results or techniques and the question connecting them. Explain which central arguments you aim to develop and how your comparison or examples will demonstrate independent understanding. State how you will check the accuracy of your account and the assumptions behind each result.                                   |
| Experimental | State the question, algorithms, baselines and measurements that you expect to use. Explain how you will check the implementation correctness, how you plan to make fair comparisons and record enough information to reproduce the work. Address repeated trials where randomness or measurement variation matters.                                  |
| Educational  | Define the audience, assumed knowledge and learning objectives. Identify the central algorithmic ideas, the resources you will create and the intended learning model. Explain how you will check the mathematical accuracy of your resources and whether the learning activities support the objectives.                                            |
| Creative     | Explain the creative intention, its intended audience, and the medium that you will use. Show how the algorithmic ideas your group plans to pursue will shape the work and what technical analysis your group will develop. Describe how tests, walkthroughs, or other suitable experiences will help assess the algorithms and intended experience. |

**Using mentoring and the notebook.** Use early mentor consultations to discuss the idea, scope, and feasibility. Bring a source, example, partial argument or outline that makes your questions concrete. Record advice and agreed next steps in your shared project notebook.

Your direction may develop after the proposal. Discuss substantial changes with your mentor, record the reasons in the notebook, and explain them at the progression review. The notebook and weekly feedback remain **unassessed**; they are not required for the proposal.

## 05 Assessment Criteria

The proposal is marked out of 20, contributing 15% of the project mark and 3% of the course mark. The criteria assess the quality of the proposed investigation and the understanding supporting it. All project formats are assessed against the same criteria, interpreted in relation to their objectives.

| Criterion                   | Marks | Evidence supporting full marks                                                                                                                                                                                                                                                                   |
|-----------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective & Motivation      | 4     | An intelligible objective, an appropriate connection to algorithms, a clear explanation of the topic's significance and purpose. Relevant assumptions and audience are identified.                                                                                                               |
| Initial background          | 5     | Relevant resources are used purposefully. A concrete and accurate explanation demonstrating initial understanding of the central themes. The proposal identifies what remains to be learned or investigated.                                                                                     |
| Scope, approach, evaluation | 5     | A substantial and manageable set of core features are established with realistic outcomes. An appropriate way of justifying or evaluating the work is demonstrated. A credible response to significant uncertainties. Any extensions labelled are distinguished from the set of core activities. |
| Feasibility & Collaboration | 5     | Concrete milestones aligned with the assessment stages, plausible timeline and resource requirements, clear next steps, and a practical plan for sharing work and reviewing contributions.                                                                                                       |
| Communication & Attribution | 2     | Clear, coherent, and concise writing. Consistent terminology and citations are clear. Appropriate acknowledgement of sources and substantial assistance.                                                                                                                                         |

Marks are awarded within each criterion for the evidence demonstrated. Partial credit recognises relevant work that is less precise, less developed or insufficiently justified; zero indicates no assessable evidence for that criterion. Completed results and claims of novelty are not needed for full marks. Explicit uncertainty can strengthen a proposal when accompanied by sound judgement about the next steps.

## 06 Sources and assistance

Cite the sources of ideas, algorithms, results, figures, and other material used or adapted. Explain your group's intended contribution, and acknowledge substantial assistance from outside the group. You remain responsible for understanding and checking the submitted material.

Follow the *Scholarship and Permitted Assistance* section of the project specification. Where AI contributes to the proposal, include a brief *Use of AI tools* statement identifying the tool and model or version where known, how it was used, and the material it contributed to. This includes generative-AI editing or rewriting. Ordinary spelling checks, formatting and reference-management functions do not require disclosure. Keep the required records of prompts, instructions and answers but they do not need to accompany this submission.